



A SURVEY ON OBJECT TRACKING IN WIRELESS SENSOR NETWORKS FOR MACHINE TOOL APPLICATIONS

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ABSTRACT

Wireless Sensor Network (WSN) applications are in increasing demand and deployed widely. Sensors communicate with other nodes through wireless Mac 802.15.4. They are low in cost, and have less power. They are used in solving the problems like surveillance, animal habitat monitoring, IoT, etc. Energy and power is limited in Sensor networks because of the fact that they are minimal in size and do data exchange between other nodes. Most of the nodes are powered using non-renewable resources. The other issues in sensors are the tracking and data aggregation. Tracking of sensor do major operations like reporting and monitoring the objects in which they are tracking inside a machine tool is a challenging task. Both these operations go hand in hand and they report to the gateway nodes which are away from the machine tool and also report to the external network. There are various algorithms on tracking is being developed over the years, mostly on the localization and prediction based. There is also cluster based method that reduces the number of sensors for tracking the objects. This paper addresses or surveys all these protocols and algorithms for object tracking for wireless sensor networks. Each algorithm is discussed in detail and finally, a comparative analysis of the object tracking algorithms in sensor networks is carried out.

Keywords: Object Tracking; Cluster and Prediction Base Object Tracking; machine tools

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1. INTRODUCTION

Wireless Sensor Network is competent solution for applications that need deep monitoring of an environment. All the nodes in the network will have a common goal wherein they work for a single operation dictated by the base station or the gateway nodes. All nodes have inbuilt intelligent so that they can communicate wirelessly with other nodes and organized in an ad hoc manner. Sensor nodes comprises of local memory, power source, smaller processing capacity, a transceiver and apart from that sensors/actuators. Wireless sensor networks are deployed at an accelerated pace and the applications of WSNs is seen in military defense, environmental hazards notification, tracking vehicles and intruders etc. Target tracking is the high-flying application where WSNs are most widely deployed to secure military regions from enemies or for wildlife monitoring. These applications have become the major use of wireless sensor networks. Based on the fact that the movements of the targets are at times predictable, Yingqi Xu et al. in [2] proposed a Prediction-based Energy Saving scheme called PES, for reducing the energy consumption to track object under acceptable conditions.

Yingqi Xu et al. in [6] presented the prediction algorithm for object tracking that minimizes power that attains power savings by keeping all the sensor nodes in sleep mode and avoiding of long-range transmissions. Also Yingqi Xu et al. in [3] proposes a reporting mechanism based on the dual prediction, in which all the nodes in the network should predict the location of the targets that are moving. All the nodes here means the base station nodes and the sensor nodes.

Kung H.T in [4] explained a publish-and-subscribe tracking method, called Scalable Tracking Using Networked Sensors (STUN) that tracks a mobile object using a network of sensors with communication set between them by using hierarchy tree. Hua-Wen Tsai et al. in [8] concentrated on how target tracks can be queried so that the mobile user can identify efficient target position. An object position can be obtained by the mobile source without sending a broadcast query. The source node usually moves towards the target when it knows the location of the target. Only with the help of the sensor nodes, the user can track and detect the mobile object when there is mobility.

2. OBJECT TRACKING METHODOLOGIES

Object Tracking Sensor Network (OTSN) consists of multiple sensor nodes deployed over a *vast monitored region* with well-defined physical boundaries [1]. The OTSN operates by placing itself between the applications and the sink nodes where it obeys the commands of the sink node that wanted the sensed data to be collected and stored. The sensed data could be temperature, pressure, etc. Also they are responsible for tracking and detecting the intruded objects which are roaming or entering their region. These sensors reports to the base station using a separate frequency that satisfies by all other sensor nodes of the network. In order to determine the states of tracked objects many sensor nodes need to work together. The following sections of the paper present various tracking methodologies basically classified into two categories namely centralized (prediction base, time base) and decentralized object tracking methodologies.

3. CENTRALIZED OBJECT TRACKING

A. Prediction Base Techniques

1) Prediction-Based Energy Saving Scheme(PES)

The sensor nodes involved is very minimal in prediction base energy saving scheme. The main objective of this algorithm is not to update the reading of a sensor node on a given time

basis [2]. This method works well if one can accept diminutive quantity of errors in predictions and modest latency in generating prediction models [4]. PES consists of three parts, a wake up technique, prediction model and a process of recovery mechanism. Current node and target nodes are used for prediction of objects where the current node predicts the locations of the target nodes. The current node when in idle goes to the sleep or idle state so that it won't predict anything. The current node is inactivated once a wakeup call is sent to the target nodes. The current node prediction works on a principle of sensing for a particular time and the sensed data is reported back to the gateway node. The current node detects the object before going to idle state and suggests that the object is just predicted within (T-X) seconds. Sleeping for the same time again, the current node and ingress nodes wake up manually to detect the object again. The above described process is repeated by the new current node, whereas further nodes can go to sleep usually.

B) Prediction model:

In [2], Yingqi Xu et al. proposed a simple prediction model assumes that the object movement is static and no movement is being predicted for a certain amount of time. The prediction results are computed based on the direction and speed used in the estimation process. Here is the heuristics for selecting the direction and speed that are prevalent in the prediction models used in the system.

- **Heuristics INSTANT:** For (t-n) seconds, the tracking objects will stay in the same speed and direction as earlier.
- **Heuristics AVERAGE:** The current node measures the speed and direction for (t-n0) seconds which was earlier stored in the mobility history. There is a communication cost involved as overhead because of the mobility among the sensor nodes.
- **Heuristics EXP AVG:** The mobility history is assigned the weights of the various stages of history and then the average value among the history is predicted.

C) Wake up mechanisms:

Target nodes are waken up to track the object, rather than waking the destination nodes up. In this case, predictions errors are calculated. Yingqi Xu et al. in [2] anticipated a wake up mechanism that chooses the membership of target nodes depending on different levels of protection.

- **Heuristic DESTINATION:** The destination node gets notification about the status of the current node.
- **Heuristic ROUTE:** From source to the destination node, the target node also placed along the route.
- **Heuristic ALL NBR:** The current node notifies the nearest nodes, en-route nodes and the destination nodes about the nodes that are surrounding to it.

D) Recovery mechanisms:

When the object is missed while tracking, the current node just send signals to all the nearest nodes about the projected route of the object as per the heuristic ALL NBR given. Also this conserves some amount of energy. This heuristics does not handle the activated nodes to find the tracking object since it is already missed. So a new approach is taken place here. If all the heuristics recovery fails, the current node performs *flooding recovery* where all the nodes are woken up in the network for object repositioning, which guarantees 0% missing rate.

2) Dual prediction-based reporting for OTSNs

The fundamental design for tracking the objects is dual prediction which makes the sensor nodes and the head nodes predict the next stage of objects. The algorithm1 shown below predicts the future tracked object mobility which was taken action at the sensor nodes. No update of information from the sensor nodes about the location of the cluster head have to be reported unless it is different from predicting the object. Also, cluster heads will not send prediction values to sensor nodes. The performance metrics like power is very less for transmitting to a nearest sensor node and it also tells the new direction of movement of a tracked object.

Algorithm: Sensor Node Prediction algorithm

Message from Sensors: *Hit_Msg (Hit)*

Variables: *S_R, P*

Functions: *Pred ()*

Procedure:

1. {When the object enters the zone of detection, the movement is predicted by the sensor based on the history of its location.}
2. $P \leftarrow \text{Pred}(\text{Hit})$
3. **while** object is within the object detection zone **do**
4. Track the object and record the readings from the sensor to *S_R*
5. **end while**
6. **if** ($S_R \neq P$) **then**
7. Send *Update_Msg (S_R)* to cluster head
8. **end if**
9. {uses previous history of the movement of the object}
10. $\text{Hit} \leftarrow (S_R, \text{Hit})$
11. Send *Hit_Msg(Hit)* to destination node

3) LR-based heuristic method

Frank Yeong-Sung Lin et al. in [10] proposes an algorithm based on heuristics for efficient and effective tracking of object in WSNs. Its based on the frequency of mobility for a given pair of sensors. They formulated this as OFF/ON integer problem. In order to solve the optimization problem, a LR-based algorithm based on heuristics is developed by Frank Yeong-Sung Lin et al in [10]. This algorithm seems to be an efficient algorithm for tracking the object almost with near optimization. Also it is scalable and faster in arriving at the solution.

$$Z_{IP} = \min \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} t_{xy}^{xy(i,j)} r_{xy}^a(i,j)$$

The purpose of the above function is to reduce the communication cost based on:

C1: It states that only one path is used from sensor node *s* to *sink* node.

C2: Any node's outgoing link to communicating node is equal to 1.

C3: If the path, *p* is chosen and the (i,j) where (i,j) forms a link) is over the path, then $Z^s_{(i,j)}$ must be equal to 1.

C4-C5: If $Z^x_{(i,j)}=0 \cup Z^y_{(i,j)}=1$, then the link (i,j) is used for repoting object's location.

C6: $\sum t^{xy}_{(i,j)}$ should be higher than or equal to 1. Constraint (7-9): Decision variables should be equal to 0 or 1.

A) Lagrangean Relaxation

They transformed the given problem into the LR problem where all the constraints are relaxed. A Lagrangean Relaxation problem for a vector of non-negative Lagrangean multipliers is given by,

Objective Function:

$$\begin{aligned} Z_{LR} (u^1_{s(i,j)}, u^2_{xy(i,j)}, u^3_{xy(i,j)}) = \min \{ & \\ \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} t^{xy}_{(i,j)} r_{xy(i,j)} a_{(i,j)} & \\ + \sum_{x \in S} \sum_{(i,j) \in L} \sum_{p \in P} u^1_{s(i,j)} x_p \delta_{p(i,j)} - & \\ \sum_{x \in S} \sum_{(i,j) \in L} u^1_{s(i,j)} z^s & \\ + \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} u^2_{xy(i,j)} 2t^{xy}_{(i,j)} - & \\ \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} u^2_{xy(i,j)} (z^y_{(i,j)} - z^x_{(i,j)} + 1) & \\ + \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} u^3_{xy(i,j)} (z^y_{(i,j)} - z^x_{(i,j)} + 1) - & \\ \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} u^3_{xy(i,j)} (t^{xy}_{(i,j)} + 1) \} & \end{aligned}$$

Here $u^1_{s(i,j)}$, $u^2_{xy(i,j)}$ and $u^3_{xy(i,j)}$ are called the Lagrangean multipliers and they are always ≥ 0 . They decomposed problem (LR) into four independent and simply solvable optimization sub-problems as follows,

$$Z_{LR} = Z_{sub1} + Z_{sub2} + Z_{sub3} + Z_{sub4}$$

Sub-Problem 1

$$\begin{aligned} Z_{sub1} (u^2_{xy(i,j)}, u^3_{xy(i,j)}) = \min & \\ \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} t^{xy}_{(i,j)} (r_{xy} a_{(i,j)} + 2u^2_{xy(i,j)} - u^3_{xy(i,j)}) & \end{aligned}$$

Sub-Problem 2

$$Z_{sub2} (u^1_{s(i,j)}) = \min \sum_{x \in S} \sum_{(i,j) \in L} (u^1_{s(i,j)} x_p \delta_{p(i,j)})$$

Sub-Problem 3

$$\begin{aligned} Z_{sub3} (u^1_{s(i,j)}, u^2_{xy(i,j)}, u^3_{xy(i,j)}) = \min \sum_{x \in S} \sum_{(i,j) \in L} & \\ [-u^3_{xy(i,j)} + \sum_{x \in S} (u^3_{xy(i,j)} - u^2_{xy(i,j)}) - \sum_{y \in S} (u^3_{xy(i,j)} - & \\ u^2_{xy(i,j)})] z^s_{(i,j)} & \end{aligned}$$

Sub-Problem 4

$$Z_{sub4} (u^2_{xy(i,j)}) = - \sum_{x \in S} \sum_{y \in S} \sum_{(i,j) \in L} u^2_{xy(i,j)}$$

A set of decision variables from Lagrangean dual problem is obtained and a LR-based algorithm is developed to adjust these local variables [10]. This algorithm gives a feasible set of solution. Frank Yeong-Sung Lin et al. stated that the upper bound and the lower bound feasible solution is guaranteed based on the optimization problem and the dual problem. When the problems are solved, the upper bound is obtained by solving the feasible solution and the lower bound is achieved by solving the dual problem. The optimality of the solution is given by duality gap computed as $(UB - LB) / LB * 100$. Smaller duality gap values yields better optimal solutions.

A) Tree Base Techniques

1) Scalable Tracking using networked sensors

This method uses a hierarchical tree method that transmits data between the sensors [4]. The nodes are named as communication nodes except that there are some nodes which are called leaf nodes which are intermediate to all the other nodes. Intermediate nodes store the information about the presence of detected objects. Leaf nodes detect object within the range of the sensors whereas the root node detects all the objects presents in the area.

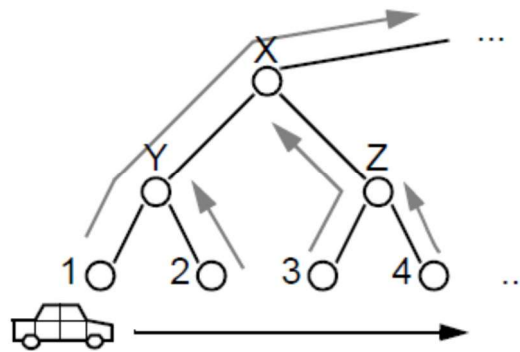


Figure 1 Scalable Tracking

Communication Cost: There will be a communication costs incurred during the detection of objects and to keep the intermediate nodes upto date about the location of the nodes. All detection messages, activated by object movements within the network are forwarded to the root. Thus, these events bring on cost of messages per unit of time.

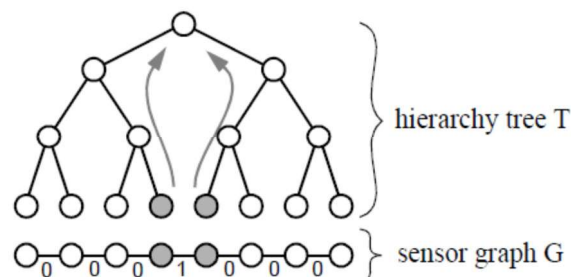


Figure 2 Messages traveling six links

This explained in brief with an example in Fig 3 that the detection event occurs anywhere between the central two sensors at unit time. The messages that were detected have to traverse to many links (6 links in this case) until they arrive at an ancestor that costs just no of messages per unit time (6 msgs/ time) These links contain only the path in between the two designated sensors. Hence the total communication cost is equal to the individual contributions of all set of adjacent sensors in Graph G as shown below,

Communication Cost (G, T) =

$$\sum_{(x,y) \in E} w(x, y) \cdot \text{PathCost}_T(i, j)$$

Low communication costs are due to the closeness of the intermediate nodes near to the root nodes that are linking the low rate edges to the other sensors [4]. For example, Fig 2 shows the hierarchy tree that have a lower communication cost because of the adjacent sensors that shares information to the intermediate nodes, rather than the root nodes. This is shown in the following Figure 3.

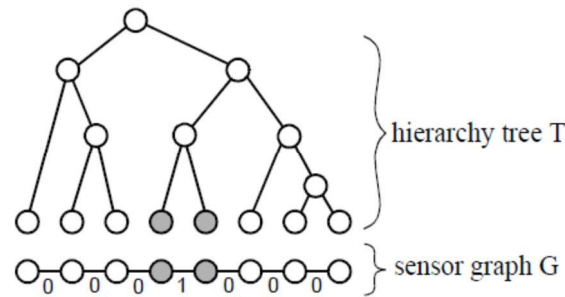


Figure 3 Sensors sharing low level intermediate node

1) Mobile Object Tracking

This Object following helps the Source (S) get the objective area and pursues the objective. The source S can't have the capacity to anticipate the moving track of target. Hua-Wen Tsai et al. work incorporates a convention that spares sensor vitality furthermore tracks target accurately quick. Firstly, S issues a flooding solicitation to get the objective position and asks sensor system to track target. Assume that S gets the objective's position is close sensor n_1 . At that point, S moves to sensor n_1 and n_1 is questioned for the objective present position. As the objective development is irregular, the objective tracks ought to be recorded by the sensors. Hua-Wen Tsai et al. in [8] recognizes distinctive zones in sensor organize by utilizing face structure. At the point when a sensor n detects an objective, it goes into its adjoining face and n is the nearest sensor to the objective and the sensor is called as entrance hub. There exists, k entrance hubs when followed target travels through k confronts in the sensor arrange. In the wake of achieving n_1 , the source S asks n_1 the following position n_2 . As the entrance hub sources in pursuing the protest, it is additionally called as reference point hub.

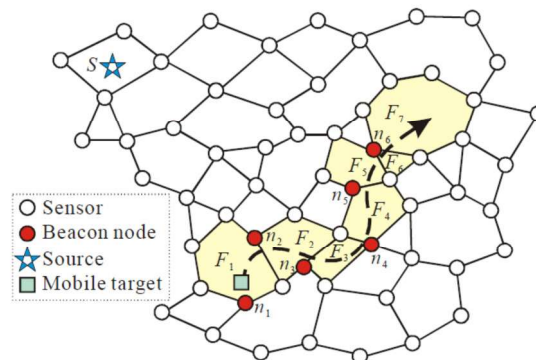


Figure 4 Neighboring faces and packet flow

a) Face Discovery:

Neighboring information is gathered by each node with a hello message and *face-neighbor nodes* are computed by Gabriel Graph. This information about the face nodes is passed by every node to its neighbors. Face neighbor nodes are decided by each node after these processes.

b) Target Discovery:

Usually a source node wants to detect an object and it needs a sensor network to track it. It sends flooded packets to the sensor networks to detect the object. When a request packet is sent to sensor n_i , it judges whether it is a *near-node*. And if a sensor n_i is the near-node and the object is in the sensing range, a reply packet is sent to source S. As every node has collected its neighbor's position in discovery phase, decision can be made whether it is nearer

to the target or not. Once the reply is sent to the source, the sensor sets itself the state as *active* and it begins to track the object as a beacon node.

c) Target Detection:

In order to track the target accurately, sensor n seeks the cooperation of its spatial neighbors. It issues *wakeup* packet to awaken the nodes of closest faces and anticipates them to track the target. After receiving wakeup packet from the sensor n , all spatial neighbors change to *active* state and detect the target simultaneously [14]. A node will be called as a near-node then it can detect the object when the object is in the tracking range of the target node and it uses the detection phase.

Hence, sensor n stops will not send anymore *wakeup* packets, but sensor m sends *wakeup* packet to its all nearby nodes. Sensors enter into sleep state when the time has expired. Thus, the *wakeup* packet wakes up the sensors in sleep state and reassign the active time of node. The node that uses beacon has to stay in *active* state waiting for the source node S .

4. DECENTRALIZED OBJECT TRACKING

A) Hidden Markov Model using Binary Sensors

This method uses binary sensors to detect the object and the binary sensors will only account if an object is detected or not [7]. In this method, sensor location is not needed and object paths are calculated based on statistical analysis. A sensor graph is formed when the links for adjacent nodes are connected. This graph forms Hidden Markov Model (HMM) which is used to calculate probable object tracks as shown in figure5.

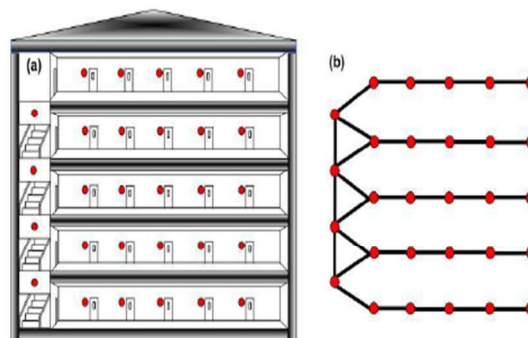


Figure 5 Hidden Markov Model

Each node contains 3 values necessary for path calculation, probability that an object starting at that node, probability that objects will be detected, and matrix of probabilities for transition to another node. In HMM, communication is in between the nodes rather than flooding the network [7].

5. COMPARATIVE ANALYSIS

The following tables illustrate how impacting factors are varying across the different object tracking techniques. Table 1 provides a comparison between centralized and decentralized techniques. Similarly Table 2 provides comparison between sensing based, collaborative signal processing [10] and prediction based techniques.

Table 1 Centralized vs. Decentralized

Comparison	Centralized Techniques	Decentralized Techniques
Cost	Expensive	Inexpensive
Power Consuming	Power Consumed	Power Efficient
Accuracy	75%	75-90%
Special Hardware	Not required	Yes
Deploy ability	Hard to deploy	Easy to deploy

Table 2 Sensing, CSP, Prediction Techniques

Comparison	Sensing Based Techniques	CSP Techniques	Prediction Based Techniques
Cost	Expensive	Inexpensive	Inexpensive
Power Consuming	Power Consumed	Power Efficient	Power Efficient
Accuracy	90%	70-85%	90%
Special Hardware	Yes	Not required	Not required
Deploy Ability	Hard	Easy	Easy

6. CONCLUSION

Object tracking in sensor networks has been the major research area in WSNs for the past. In our work, we investigated the various object tracking techniques evolved over the years and came out with their pros and cons in the comparative analysis. The selection of algorithm/technique depends on type of application or the conservative aspects we are concentrating on. For example, prediction based techniques yields better results in terms of energy consumption as they predict future movement of the objects and employ certain wake up mechanisms that wakes up only few sensors in the network. Likewise, if a user want to chase a target when both user and target moving at different speeds, mobile object tracking technique described in this previous sections can be utilized. Each technique emphasizes on different factors like ease of deployment, accuracy, power consumption, inexpensiveness etc.

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